

GF(256) In Class

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Irreducible Polynomial for GF(256)

$$= x^8 + x^4 + x^3 + x^2 + 1$$

Full 256-element field-to-binary table can be found in the exported pdf file.

Multiple Addition Examples

1. $a = 58$ $b = 41$

$$\begin{array}{r} \oplus \quad 00111010 \\ \quad 00101001 \\ \hline 00010011 \end{array} \rightarrow \boxed{00010011}$$

3. $a = 4$ $b = 641$

$$\begin{array}{r} \oplus \quad 00000100 \\ \quad 010001000 \\ \hline 01000100 \end{array} \rightarrow \boxed{01000100}$$

2. $a = 37$ $b = 85$

$$\begin{array}{r} \oplus \quad 00100101 \\ \quad 01010101 \\ \hline 01110000 \end{array} \rightarrow \boxed{01110000}$$

Multiple Multiplication Examples

1. $a = 58$ $b = 41$

$$58 = 00111010 \rightarrow x^5 + x^4 + x^3 + x$$

$$41 = 00101001 \rightarrow x^5 + x^3 + 1$$

Reduce using irreducible

$$p(x) = x^8 + x^4 + x^3 + x^2 + 1$$

$$x^{10} \div x^8 = x^2 \rightarrow x^2(x^8 + x^4 + x^3 + x^2 + 1)$$

$$x^{10} + x^6 + x^5 + x^3 + x$$

$$x^{10} + x^6 + x^5 + x^4 + x^2$$

$$(x^5 + x^4 + x^3 + x)(x^5 + x^3 + 1)$$

$$x^{10} + x^8 + x^5 + x^9 + x^7 + x^4 + x^8 + x^6 + x^3 + x^6 + x^4 + x$$

$$x^{10} + x^9 + x^7 + x^5 + x^3 + x$$

Reduce Again!

$$x^9 \div x^8 = x \rightarrow x(x^8 + x^4 + x^3 + x^2 + 1)$$

$$x^9 + x^7 + x^6 + x^4 + x^5 + x^2 + x = x^7 + x^6 + x^5 + x^2$$

$$x^9 + x^5 + x^4 + x^3 + x$$

$$x^7 + x^6 + x^5 + x^2 = 11100100 = 2^7 + 2^6 + 2^5 + 2^2$$

$$= 228$$

Other Examples are in Program

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Division Examples

$$a = 4, b = 64$$

$$4 \rightarrow 00000100 \rightarrow \alpha^2 \quad 4 = \alpha^2 = 4(64^{-1})$$

$$64 \rightarrow 01000000 \rightarrow \alpha^6 \quad 64 = \alpha^6 \quad (\alpha^6)^{-1} = \alpha^{-6}$$

$$2^{255} \text{ irreducible polynomial} = x^{255} + x^{233} + x^{231} + 1$$

$$-6 \bmod 255 = 249$$

$$64^{-1} = \alpha^{249}$$

$$4(64^{-1}) = \alpha^2 \alpha^{249} = \alpha^{251} = \alpha^{251-255} = \alpha^{-4} = \alpha^{251}$$

From Program

Checking Work

Addition: The addition result agrees with the $GF(256)$ arithmetic developed in class because addition is performed using bitwise XOR.

Multiplication: The multiplication result agrees with what we developed in class. The binary values are multiplied as polynomials using binary arithmetic and then are reduced by the irreducible polynomial.

Division: The division result agrees with what we developed in class. The field elements can be written as powers of α . Division is then performed by subtracting the exponents mod 255.

What we learned about how our programming language supports $GF(256)$

We used the programming language R in RStudio. We learned that R can be used to implement $GF(256)$ by using binary operations and functions. R does not perform the arithmetic automatically. We created functions for addition, multiplication, division, and inverses using the rules learned in class.